UWB–IMU–Odometer Fusion for Simultaneous Calibration and Localization

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***Abstract*—The location accuracy of fixed anchors plays a pivotal role in ultrawideband (UWB) positioning. However, existing calibration methods for calculating the anchor positions require anchor-to-anchor communication or known initial values for anchor locations. Moreover, the calibration accuracy is adversely affected by non line-of-sight (NLOS) conditions. We propose a two-stage calibration scheme to conduct simultaneous calibration and localization (SCAL) based on UWB–inertial measurement unit (IMU)–odometer sensor fusion without the need for anchor-to-anchor communication or manual intervention. Our method performs IMU–odometer-aided multidimensional scaling (IO-MDS) to provide the initial calibration value without anchor-to-anchor ranging measurements. This is followed by a novel factor-graph-based framework to achieve coarse-to-fine calibration based on UWB, inertial, and odometer measurements. In existing works, a single UWB range measurement is regarded as a weak constraint, as it often leads to incorrect estimates. Our method uses the derived radial velocity (DRV) and IO-MDS factors as additional strong constraints for reliable estimates. To minimize the NLOS influence on the calibration process, we introduce an improved least square-support vector machine (ILS-SVM) based on adaptive weight parameter and multikernel function. Experimental results on two field collected datasets show enhancements in NLOS identification and SCAL. In the lab dataset, identification accuracy increased by 5.5%, with improvements of 0.124 m and 0.309 m in root mean square error (RMSE) for robot and anchor locations, respectively. In the parking lot dataset, identification accuracy improved by 5.0%, with RMSE improvements of 0.223 m and 0.317 m for robot and anchor locations, respectively.**

***Index Terms*—Calibration, factor graph, localization, non line-of-sight (NLOS), ultrawideband (UWB)/inertial measurement unit (IMU)/odometer.**

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I. Introduction

LOCATION awareness systems are widely used in everyday life, e.g., for automatic picking tasks, automated guided vehicles (AGVs), automated valet parking (AVP), and object tracking [1]. Ultrawideband (UWB) is considered as one of the most promising technologies in next-generation wireless sensor networks (WSNs) since it achieves cm-level ranging accuracy [2], [3].

By measuring distances to static modules with known positions (called anchors), a localization system can estimate the position of the mobile node (the tag) [4]. Imagine a system that relies on multiple anchor nodes at fixed positions to localize a mobile tag. In that case, a primary calibration procedure is to manually or automatically measure the exact location of the installed anchors. Conventional calibration can be performed by means of dedicated surveying and mapping techniques, such as using RTK-GPS or laser range finding, to measure all anchor-to-anchor distances and determine every anchor's coordinates. Nevertheless, surveying the anchors is typically a labor-intensive, time-consuming, and error-prone process that requires human intervention. Furthermore, precalibration process is challenging to perform in narrow underground spaces and caves as they are difficult to reach and dangerous, and in high-radiation areas due to interference.

To overcome these challenges, self-calibration mechanisms are employed to automatically detect anchors and compute their positions during the deployment of the UWB network. Based on UWB communication modes, the ranging measurements are often divided into two main categories: 1) anchor-to-anchor ranging and 2) anchor-to-tag ranging [5]. Implementing these ranging distances, the self-calibration procedure can be generally achieved automatically without manual intervention. A typical anchor calibration process first determines the time-of-arrival distance between each other, and then calculates the anchor's physical location based on the multidimensional scaling (MDS) algorithm [6]. However, these UWB calibration solutions require interanchor measurements and no occlusion between the UWB anchors. In practical applications, many UWB manufacturers do not provide anchor-to-anchor communication interfaces to users,

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i.e., Decawave,[1] Nooploop,[2] etc. The system must first find the tag locations when only anchor-tag ranges are available. The calibration process then boils down to a reverse localization problem that can be solved with ranging distances between the anchors and the moving tag. However, a good initial guess of tag locations is challenging.

Additionally, optimizing anchors and tag position information simultaneously is similar to the simultaneous localization and mapping (SLAM) problem, the only difference being that the main focus is calibration instead of mapping. Constructing robust constraints in a graph framework for finding an optimal solution is also challenging.

The accuracy of self-calibration methods relies heavily on the ranging capabilities of UWB. The calibration accuracy degrades when anchors are obstructed by physical barriers, such as walls or metal structures, causing non line-of-sight (NLOS) conditions and introducing nonnegligible positive biases in ToF-based range measurements. Thus, it becomes crucial to identify line-of-sight (LOS)/NLOS conditions and remove the associated errors. Existing NLOS identification methods mainly comprise distance-statistics-based methods, channel-character-based methods, and fusion-based methods [7], [8], which divide the measurements into identified-NLOS and identified-LOS. Machine learning (ML) methods have proven to be suitable tools for addressing statistical analysis and data characteristic problems. As such, support vector machine (SVM) [9], [10] and decision trees [11] have been proposed to improve the identification accuracy of LOS/NLOS conditions. However, these methods rely solely on UWB ranging or received signal strength (RSS) measurements, resulting in sensitivity to outliers and noises with a nonnormal distributions, thus limiting their adaptability across various practical scenarios. Based on this, the problem of LOS/NLOS identification remains challenging in large-scale-UWB-based localization systems deployed in complex indoor environments, which are characterized by mixed LOS/NLOS propagation conditions created by multiple anchors and obstacles.

In this article, we explore the combination of UWB, inertial, and odometer sensors to simultaneously estimate the unknown positions of the robot and anchors based on the assumption that only anchor-to-tag communication is available. The main contributions of this article are as follows.

1) An improved least squares version of the SVM (ILS-SVM) is designed based on UWB RSS and ranging measurements, multidimensional characteristics of UWB signals are utilized to enhance the NLOS identification performance. Combining the adaptive weight parameter and the multikernel function, thereby further improving the accuracy and robustness of LOS/NLOS identification.
2) A two-stage calibration scheme is proposed to achieve coarse-to-fine anchor location estimation without anchor-to-anchor ranging or initial location assumptions. Classical UWB anchor calibration methods typically require measuring distances between anchor pairs or rely on known initial values for anchor locations. However, many UWB manufacturers do not provide anchor-to-anchor communication interfaces, and obtaining initial anchor locations can be challenging. To enhance the applicability of UWB anchor calibration, we propose an extension to the MDS algorithm in the first stage, called inertial-measurement-unit-(IMU)–odometer-aided MDS (IO-MDS), which uses IMU–odometer trajectory constraints between tag positions. In the second stage, our goal extends beyond merely treating each sensor's measurements as a factor. We propose to include the global position constraints from IO-MDS and the relative motion constraints from derived radial velocity (DRV) within factor graphs to achieve coarse-to-fine pose estimation. To the best of our knowledge, this is the first work to propose a two-stage processing mechanism for simultaneously locating the tag and anchors without requiring anchor-to-anchor ranging or prior location information.
3) The proposed ILS-SVM method and two-stage calibration scheme are applied to a UWB–IMU–odometer-based localization system. Multiple experiments are conducted under lab and parking lot scenarios in LOS/NLOS conditions. The results show the superiority of the proposed method compared

TABLE I
MATHEMATICAL NOTATIONS

Symbols	Definitions
RSS	Difference between the relative power of the received signal and the power of the first detected signal.
$\boldsymbol{\omega}$	Weight matrix in LS-SVM model.
b	Bias term in LS-SVM model.
$\varphi(\cdot)$	Nonlinear function that maps input data to a high-dimensional feature space.
P_i	RSS received from the ith UWB anchor.
r_i	Range measurement received from the ith UWB anchor.
$n(\mathbf{b}(\cdot))$	Number of samples within a sliding window.
γ_i	Adaptive weight of the ith UWB sample point.
$\mathcal{N}(\cdot)$	Z-score normalized function.
$H(\cdot)$	Composite kernel function.
$\mathbf{X}_0$	Sequence of tag's positions in physical space.
$\mathbf{x}_{t_i}$	The ith position of tag in physical space.
$\mathbf{X}_1$	Sequence of anchors' positions in physical space.
$\mathbf{x}_{a_j}$	Position of the jth anchor in physical space.
d_{t_i,a_j}	Estimated Distance between the ith tag position and the jth anchor.
$f(\cdot)$	MDS cost function.
$\tilde{\mathbf{X}}_0$	Positions of the target tag in signal space.
$\tilde{\mathbf{X}}_1$	Positions of anchors in signal space.
$\mathbf{T}$	Transformation matrix from signal space into physical space.
$\mathbf{D}$	Distance matrix.
r_{t_i,a_j}	UWB TOF ranging measurement distance between the ith tag position and the jth anchor.
δ_{t_i,t_j}	Derived relative position from IMU-odometer system between tag position x_{t_i} and x_{t_j}.
P_{t_i,a_j}	RSS value between the ith tag position and the jth anchor.
$\tilde{\rho}_{t_i,a_j}$	Weight of the UWB range measurement between the ith tag position and the jth anchor.
$o(\cdot)$	Error model of the IMU-odometer system.
$\tilde{\gamma}_{t_i,t_j}$	Weight of the measurements between the ith tag position and the jth tag position from IMU-odometer system.
l_{t_i,t_j}	Distance between the ith and the jth tag position.
$\xi(\cdot)$	Pre-integration of IMU measurements.
$\tau(\cdot)$	Pre-integration of odometer measurements.

[1]https://www.qorvo.com/products/p/DWM3000EVB
[2]Nooploop. LinkTrack User Manual, v2.1 edition.

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TABLE II
OVERVIEW OF SELF-CALIBRATION AND LOCALIZATION SOLUTIONS FOR UWB NODES

Solution	Conditions	Sensors	Nodes	Area	Communication mode	Ranging mode	Calibration Accuracy	Localization accuracy	NLOS identification
MLE [5]	LOS	UWB	4	17×6 m^2	Anchor-tag	ToF	10cm	-	-
Gauss-Newton [13]	LOS	UWB	8	22×15 m^2	Anchor-anchor	ToF and TDOA	16 cm	-	-
BFGS and KF [14]	LOS/NLOS	UWB	18	25×11 m^2	Anchor-anchor and Anchor-tag	TWR	23.3 cm	-	-
ESKF [15]	LOS/NLOS	UWB and IMU	5	20×20 m^2	Anchor-tag	TWR	9.05 cm	5.59 cm	✓
PF [16]	LOS/NLOS	UWB and IMU	4	30×2 m^2	Anchor-tag	ToF	15.9 cm	18.9 cm	✓
MGBP [17]	LOS/NLOS	UWB and IMU/odometer	7	30×80 m^2	Anchor-tag	ToF	94 cm	110 cm	✓
IPFHO [18]	-	UWB	4	10×10 m^2	Anchor-tag	TDOA	-	21 cm	-
VB-CSRUKF-RTS [19]	LOS/NLOS	UWB	4	-	Anchor-tag	ToF	10.58 cm	39.7 cm	✓

to existing methods. To benefit the community, the relevant codes and datasets of this work are released at https://github.com/JasonSun623/SCAL.

The remainder of this article is organized as follows. In Section II, existing solutions for UWB anchor self-calibration and LOS/NLOS identification are reviewed. Section III gives details of the proposed UWB NLOS identification and the two-stage calibration methods. To validate the performance of the aforementioned methods, extensive experiments are conducted in Section IV. Section V concludes this article. The meanings of the mathematical notations are listed in Table I.

II. RELATED WORK

In the existing studies [6], [12], the majority of the calibration schemes for WSN focus on Wi-Fi or RFID technologies, often under LOS or simulated conditions, rather than real-world deployments incorporating both LOS and NLOS conditions. Besides, the conventional UWB self-calibration solutions [13], [14] focus on exploiting anchor-to-anchor ranging or angle measurements, rather than solely relying on anchor-to-tag ranging. In this section, we first discuss UWB self-calibration based on different UWB communication modes and then provide an overview of UWB LOS/NLOS identification solutions. In Table II, we summarize the most relevant works on UWB self-calibration and localization.

A. UWB Self-Calibration and Localization

In this section, we discuss solutions that are based on UWB communication measurements and do not require manual intervention. UWB systems that allow anchor-to-anchor communication in the network, compute the position of the uncalibrated anchors from the distances between all anchors. Piavanini et al. [13] proposed an iterative approach that optimizes the minimum distance error between true and estimated distances by exploiting anchor-to-anchor ranging measurements. Experimental results showed a root mean square error (RMSE) of 16 cm. Herbruggen et al. [14] proposed a three-stage calibration algorithm for LOS/NLOS mixed environments. The initialization stage employs a heuristic approach to establish initial positions for all anchors. Subsequently, the global optimization stage utilizes the Broyden–Fletcher–Goldfarb–Shanno (BFGS) algorithm to correct multihop errors accumulated during the initialization stage. Finally, an optional extra calibration stage updates anchor positions using a Kalman filter. However, this algorithm requires at least three anchor coordinates to be available for initialization. When only anchor-tag communication is available, the anchor calibration method generally utilizes additional motion sensors and anchor-tag ranging to achieve anchor position estimation. Inspired by the SLAM technique, a tightly integrated fusion approach is presented in [15], combining UWB and low-cost IMU sensors. This approach employs an error-state Kalman filter (ESKF) to simultaneously estimate the mobile tag's position and reconstruct the spatial layout of anchors. Wang et al. [16] introduced RO-SLAM, which utilizes a region-based particle filter and a proper likelihood model. More specifically, the region-based particle filter initializes the locations of the landmarks (anchors). To deal with the introduction of new landmarks, a proper likelihood model is designed without delayed processing. Xiong et al. [17] designed a multihypothesis Gaussian belief propagation (MGBP) positioning algorithm that combines characteristics of multihypothesis filtering and factor graph optimization to initialize the estimated anchor states without relying on an external estimator. The factor graph incorporates UWB ranging and odometer measurements as constraints, leveraging the benefits of multiple hypothesis filters to represent anchor uncertainties. Furthermore, MGBP exhibits the capability to impose constraints on anchor positions even when bearing information is unavailable. However, the effectiveness of MGBP is intricately linked to the accuracy of the nonlinear ranging model. Thus, deviations in the estimation of model parameters could compromise the reliability of inference outcomes. Luo et al. [18] proposed an improved particle-filter-based hybrid optimization (IPFHO) algorithm to reduce the influence of the uncertain measurements and address large positioning errors. This algorithm combines the adaptive particle filter and search optimization to handle ranging errors and anchor node calibration errors. However, this algorithm requires initial coordinates of all anchor nodes. Cao et al. [19] integrated the variational Bayesian (VB), constrained square root-unscented Kalman filter (CSRUKF) and robust Taylor series (RTS) methods (VB-CSRUKF-RTS) based on calibration to improve the UWB positioning accuracy in harsh underground environments. Anchor calibration method is employed with the aid of reference nodes (RNs) under LOS conditions. Then, the VB-CSRUKF strategy is

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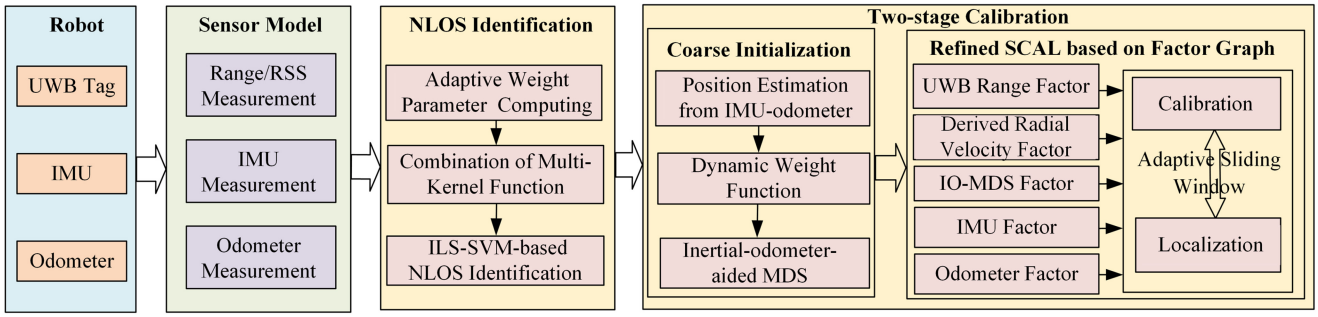


Fig. 1. Overall system architecture. Our system processes the inputs from UWB tag, IMU, and odometer sensors mounted on a mobile robot. We propose an ILS-SVM-based method to perform NLOS detection based on the RSS and ranging information. Next, we design a two-stage automatic calibration scheme that does not rely on anchor-to-anchor communication or known initial values for anchor locations. The first stage adopts the proposed IO-MDS method that obtains initial anchor position estimates. The second stage simultaneously refines anchor positions and robot trajectories by means of the FGO framework involving the strong constraints of the DRV and IO-MDS outcomes.

designed to cope with NLOS measurement noise. However, this method requires appropriate RNs locations in the process of calibration.

B. UWB LOS/NLOS Identification

NLOS identification mainly relies on ML-based methods to differentiate between LOS and NLOS conditions [20]. Various ML methods have been investigated in the literature, such as random forests (RFs) and SVMs. Sun et al. [21] developed an NLOS identification algorithm by using an RF classifier, leveraging the normalized received signal's characteristics. LOS and NLOS classification results are derived from the trained decision tree. Che et al. [22] treated the identification process as a weighted Naive Bayes (WNBs) problem, demonstrating that a probability attribute weighting scheme enhanced the accuracy of imbalanced NLOS identification in a harsh indoor environment. Barral et al. [23] employed cross-validation mode to train SVM models to identify LOS, Soft-NLOS, and Hard-NLOS scenarios, utilizing ranging distance and RSS as training data. Additionally, they introduced LS-SVM which exhibits reduced training data requirements and lower complexity. Huang et al. [8] changed the classification problem to a linear programming problem and solved it by using the Lagrange methodology. However, these methods are sensitive to outliers and noises in the UWB raw measurements, resulting in reduced recognition accuracy and robustness.

III. System Overview

Fig. 1 provides an overview of the proposed system. A mobile robot is equipped with an odometer, a 6-DoF IMU and a UWB sensor rigidly attached to the body frame, with all intrinsic and extrinsic parameters calibrated. First, we propose an ILS-SVM classifier based on UWB RSS and ranging measurements to identify LOS and NLOS situations. When the LOS measurements are detected, the coarse calibration of anchors is conducted to minimize the residual between the predicted robot positions and unknown anchor positions by the proposed IO-MDS method. After that, the rough position, inertial, and odometer measurements are seamlessly integrated into the factor graph framework to simultaneously optimize the positions of anchors and track the state of the robot.

A. UWB NLOS Identification

The precision of UWB range measurement is affected by the NLOS propagation effect and Gaussian noise, making it challenging to obtain high-accuracy calibration. We propose an ILS-SVM to solve the impact of both NLOS range errors and UWB measurement noises. In this section, the LS-SVM algorithm is briefly reviewed first. Next, we give details of our ILS-SVM design that deals with UWB range and RSS measurements.

1) LS-SVM Algorithm: LS-SVM algorithm is a practical improvement of the conventional SVM algorithm, which is good at nonlinear estimation even when the sample size is small [24], [25]. Assuming that a training set with n samples is $\{(r_i, y_i)\}_{i=1}^{n}$, r_i is the input UWB range data and $y_i \in \{-1, +1\}$ is the input label as NLOS or LOS. The regression function of the conventional LS-SVM algorithm is expressed as

$$y_i = \text{sign}\left[\boldsymbol{\omega}^T \varphi(r_i) + b\right] \tag{1}$$

where $\boldsymbol{\omega}$ is the weight matrix, b represents the bias term, $\varphi(\cdot)$ is the function to map the input data to a higher dimensional feature space, and "sign" is the signum function. For solving the estimation problem of LS-SVM, we minimize the objective function with an equality constraint that can be formulated as follows:

$$\begin{cases} \min J(\boldsymbol{\omega}, b, \psi) = \frac{1}{2}\boldsymbol{\omega}^T\boldsymbol{\omega} + \frac{1}{2}\eta \sum_{i=1}^{n} \Delta\psi_i{}^2 \\ \text{s.t. } y_i = \boldsymbol{\omega}^T \varphi(r_i) + b + \Delta\psi_i, i = 1, 2, \ldots, n \end{cases} \tag{2}$$

where $\Delta\psi_i$ means the random error term of the ith sample points and η is the user-specified regularization parameter that adjusts the complexity and accuracy of the estimated function.

2) Adaptive Weight Parameter: During preprocessing of UWB range data containing random noise, conventional LS-SVM assigns equal weight to each range error. However, UWB ranging and RSS rely on different physical principles. Each measurement type is subject to distinct errors due to uncertainties in the material and thickness of obstacles. To address this limitation, we propose adaptive weight parameter that combines RSS and ranging information in a sliding window. This enhancement improves the robust performance of LS-SVM methods by balancing the impact on the regression model. The adaptive weight parameter is then incorporated into the optimization model.

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The accuracy of the measured UWB ranging distance is directly affected by the LOS/NLOS conditions. The variance in RSS and ranging measurements can be used to identify these conditions. NLOS conditions generally result in higher variance on RSS and ranging measurements due to signal attenuation and multipath effects caused by obstacles or reflections. Based on this, we compute the variance of the RSS and ranging samples in a sliding window w

$$\mathrm{var}(P_i) = \frac{1}{n(\mathbf{b}(w))} \sum_{i\in\mathbf{b}(w)} \left(P_i - \frac{1}{n(\mathbf{b}(w))} \sum_{i\in\mathbf{b}(w)} P_i \right)^2 \tag{3}$$

$$\mathrm{var}(r_i) = \frac{1}{n(\mathbf{b}(w))} \sum_{i\in\mathbf{b}(w)} \left(r_i - \frac{1}{n(\mathbf{b}(w))} \sum_{i\in\mathbf{b}(w)} r_i \right)^2$$

where $n(\mathbf{b}(w))$ represents the number of samples within a sliding window w.

Thus, a novel adaptive weight parameter of the ith UWB sample point can be set as follows:

$$\begin{cases} \gamma_i = 1, & \mathrm{var}(x_i) \le c_1 \\ \frac{1}{\alpha|\mathcal{N}(\mathrm{var}(P_i))|} + \frac{1}{\beta|\mathcal{N}(\mathrm{var}(r_i))|}, & c_1 \le \mathrm{var}(x_i) \le c_2 \\ 0, & \text{otherwise} \end{cases} \tag{4}$$
$$x_i = \mathcal{N}(\mathrm{var}(P_i)) + \mathcal{N}(\mathrm{var}(r_i))$$

where $\mathcal{N}(\cdot)$ is the Z-score normalized function, c_1 and c_2 are range tuning constants, α and β are the corresponding positive weights, and x_i is normalized variance of the RSS and ranging samples.

Next, (2) can be solved to find the optimal value by introducing the Lagrange multiplier function as follows:

$$L(\boldsymbol{\omega}, b, \Delta\psi_i, \lambda_i) = \frac{1}{2}\boldsymbol{\omega}^T\boldsymbol{\omega} + \frac{1}{2}\eta \sum_{i=1}^{n} \gamma_i \Delta\psi_i^2 - \sum_{i=0}^{n} \lambda_i \left[\boldsymbol{\omega}^T \varphi(x_i) + b + \Delta\psi_i - y_i \right] \tag{5}$$

where λ_i is the Lagrange multiplier which denotes the importance of the respective input data x_i. Hence, assigning a higher weight γ_i to the x_i with a smaller variance of RSS and range can result in more accurate measurement results. According to Karush-Kuhn-Tucker conditions, we have

$$\begin{cases} \frac{\partial L}{\partial \boldsymbol{\omega}} = 0 \to \boldsymbol{\omega} = \sum_{i=1}^{n} \lambda_i \varphi(x_i) \\ \frac{\partial L}{\partial b} = 0 \to \sum_{i=1}^{n} \lambda_i = 0 \\ \frac{\partial L}{\partial \Delta\psi_i} = 0 \to \lambda_i = \eta\gamma\Delta\psi_i \\ \frac{\partial L}{\partial \lambda_i} = 0 \to \boldsymbol{\omega}^T\varphi(x_i) + b + \Delta\psi_i - y_i = 0. \end{cases} \tag{6}$$

Then, λ_i and b can be calculated by introducing kernel functions.

3) Multikernel Function: The choice of kernel function significantly impacts the prediction accuracy and convergence domain of an LS-SVM model [10]. Linear kernel function simplifies the computation between input vectors through a simple dot product, effectively addressing the curse of dimensionality. However, it may not adequately capture complex nonlinear relationships between input vectors. On the contrary, the radial basis kernel function (RBF) is well-suited for computing principal components based on distance in a high-dimensional feature space. Nonetheless, it can increase computational time. To balance accuracy and computational complexity in the LS-SVM method, we combine the linear kernel function and RBF, as follows:

$$H(x_i, x_j) = \mu x_i^T x_j + (1-\mu) e^{-\frac{\|x_i - x_j\|_2^2}{2\sigma^2}} \tag{7}$$

where $\|x_i - x_j\|_2^2$ is the squared Euclidean distance between the two feature vectors and σ represents the width parameter of the RBF kernel. μ is implicated in tuning the weight assigned to each kernel function.

Consequently, the decision function for regression becomes

$$y_i = \mathrm{sign}\left[\sum_{i=1}^{n} \sum_{j=0}^{n} \lambda_i H(x_i, x_j) + b \right]. \tag{8}$$

We then distinguish LOS and NLOS conditions using the trained ILS-SVM classifier as

$$\begin{cases} \boldsymbol{\omega}^T\varphi(x_i) + b \ge +1, \text{ the link is LOS} \\ \boldsymbol{\omega}^T\varphi(x_i) + b \le -1, \text{ the link is NLOS.} \end{cases} \tag{9}$$

When the link is classified as NLOS, it will be discard. Note that this identify-and-discard approach is based on our prior knowledge of the LOS/NLOS conditions in experimental settings. We ensured that at least three anchors were always within the LOS link across the entire localization space.

B. Two-Stage Calibration

In this section, we present the solution to the anchor self-calibration problem. The goal is to simultaneously estimate the positions of all unknown anchors and track the state of the tag. To derive a convergent and stable self-calibration result, we need proper initialization.

The coarse-to-fine anchor self-calibration process is divided into two stages: 1) we roughly estimate UWB anchor positions with the proposed IO-MDS method and 2) based on coarse calibration initialization results, we take the DRV factor into consideration and formulate the factor graph optimization to obtain more accurate estimation results.

Throughout the entire self-calibration process aimed at computing the UWB anchor positions in a 2-D reference frame, an IMU sensor can derive the tag's position in the 2-D reference frame from acceleration and angular velocity measurements. To provide a coarse initial guess of the anchor position, we move the mobile robot to random positions while collecting UWB ranging measurements from different UWB anchors. We can initialize the position of the anchor by extending the approach presented in [26], and IO-MDS is proposed. First, we give a brief introduction to the classical MDS and explain how to apply it to the anchor positions calibration task. Then, a detailed description of our IO-MDS is presented.

1) Multidimensional Scaling: MDS is a data processing algorithm that converts high-dimensional data into a low-dimensional space. MDS-based localization has been commonly used in WSN [6], [26]. These methods can use the similarity between pairwise connectivity information to

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compute the relative space coordinates of each node in the entire WSN. Considering N UWB anchors at unknown locations $\mathbf{X}_1 = [\mathbf{x}_{a_1}, \mathbf{x}_{a_2}, \ldots, \mathbf{x}_{a_N}]^T$, the pairwise distance between the ith anchor and the jth anchor can be defined as

$$d_{a_i,a_j} = \left\|\mathbf{x}_{a_i} - \mathbf{x}_{a_j}\right\|_2. \tag{10}$$

The MDS cost function of anchor-to-anchor position errors is given as

$$f\left(\tilde{\mathbf{X}}_1\right) = \underset{\mathbf{X}_1}{\operatorname{argmin}} \sum_{i<j} \left(r_{a_i,a_j} - d_{a_i,a_j}\right)^2 \tag{11}$$

where $\tilde{\mathbf{X}}_1$ are the coordinates of anchor nodes under the signal space, and r_{a_i,a_j} is the actual UWB TOF ranging measurement distance between anchor i and j. The SMACOF algorithm [27] is used to solve the cost function in (11) by iteratively adjusting distances between data points in the signal space.

After that, we transform the coordinates of anchors in signal space into the physical space

$$\begin{aligned} \mathbf{X}_1 &= \mathbf{T}\tilde{\mathbf{X}}_1 \\ f(\mathbf{T}) &= \min_{\mathbf{T}} \frac{1}{2} \sum \left\|\mathbf{X}_1 - \mathbf{T}\tilde{\mathbf{X}}_1\right\|_2 \end{aligned} \tag{12}$$

where $\mathbf{T}$ is the transformation matrix and $\tilde{\mathbf{X}}_1$ are the coordinates of anchor nodes under the signal space.

2) Coarse Initialization Based on IO-MDS: Considering the tag with M derived positions $\mathbf{X}_0 = [\mathbf{x}_{t_1}, \mathbf{x}_{t_2}, \ldots, \mathbf{x}_{t_M}]^T$ from the IMU–odometer system, we define the distance matrix $\mathbf{D}$ that contains relative distance measurements for all nodes as follows:

$$\mathbf{D} = \begin{bmatrix} \mathbf{D}_{11} & \mathbf{D}_{12} \\ \mathbf{D}_{21} & \mathbf{D}_{22} \end{bmatrix} \tag{13}$$

where $\mathbf{D}$ consists of four chunk matrices and $\mathbf{D}_{11}$ represents the range between UWB anchors. Due to the unavailable anchor-to-anchor ranging, $\mathbf{D}_{11}$ is a zero matrix. $\mathbf{D}_{12}$ represents the range between the tag and anchor, and $\mathbf{D}_{22}$ is the incremental position matrix of the tag through dead reckoning introduced in [28]

$$\begin{aligned} \mathbf{D}_{12} &= \begin{bmatrix} r_{t_1,a_1} & \cdots & r_{t_{M-1},a_1} & r_{t_M,a_1} \\ r_{t_1,a_2} & \cdots & r_{t_{M-1},a_2} & r_{t_M,a_2} \\ \vdots & & \vdots & \vdots \\ r_{t_1,a_N} & \cdots & r_{t_{M-1},a_N} & r_{t_M,a_N} \end{bmatrix} \\ \mathbf{D}_{22} &= \begin{bmatrix} 0 & \delta_{t_1,t_2} & \cdots & \delta_{t_1,t_M} \\ \delta_{t_2,t_1} & 0 & \cdots & \delta_{t_2,t_M} \\ \vdots & \vdots & \ddots & \vdots \\ \delta_{t_M,t_1} & \delta_{t_M,t_2} & \cdots & 0 \end{bmatrix} \\ \mathbf{D}_{21} &= \mathbf{D}_{12}^T \end{aligned} \tag{14}$$

where δ_{t_i,t_j} represents the derived relative position from the IMU–odometer system between tag position x_{t_i} and x_{t_j}.

In practice, after UWB NLOS identification, most NLOS measurements are removed, leaving only correct measurements and outliers with Gaussian noise smaller than the baseline. IMU–odometer measurements are also affected by noise. Based on this, dynamic weight functions are introduced to account for the impact of measurement noises. We represent the UWB range error by using the RSS variance

$$\operatorname{var}\left(P_{t_i,aj}\right) = \frac{1}{M \times N} \sum_{i<j} \left(P_{t_i,aj} - \frac{\sum_{i<j} P_{t_i,a_j}}{M \times N}\right)^2. \tag{15}$$

The normalized dynamic weight function of the UWB range is as follows:

$$\begin{aligned} \tilde{\rho}_{t_i,a_j} &= \frac{1}{\sqrt{2\pi}} e^{-\frac{\operatorname{var}\left(P_{t_i,a_j}\right)}{2}} \\ \rho_{t_i,a_j} &= \frac{\tilde{\rho}_{t_i,a_j}}{\sum_{i<j} \tilde{\rho}_{t_i,a_j}} \end{aligned} \tag{16}$$

where ρ_{t_i,a_j} denotes the corresponding weight of the UWB range measurement.

From [29], we can obtain a model of the IMU/odometer system and the relative position error between x_{t_i} and x_{t_j} over a short time period Δt

$$\begin{aligned} \sigma_{t_i,t_j} &= o\left(\eta^b, b^a{}_{\Delta t}, b^g{}_{\Delta t}, \eta^o, \delta x_{t_i,t_j}\right) \\ \delta x_{t_i,t_j} &= x_{t_i} - x_{t_j} \end{aligned} \tag{17}$$

where $o(\cdot)$ denotes the error model of the IMU–odometer system, and η^b and η^o are zero-mean white Gaussian noises of IMU and odometer, respectively. $b^a_{\Delta t}$ and $b^g_{\Delta t}$ are the time-varying accelerometer bias and gyroscope bias, respectively. The dynamic weight function of the IMU–odometer system can be expressed as

$$\begin{aligned} \tilde{\gamma}_{t_i,t_j} &= \frac{1}{\sqrt{2\pi}} e^{-\frac{\sigma^2_{t_i,t_j}}{2}} \\ \gamma_{t_i,t_j} &= \frac{\tilde{\gamma}_{t_i,t_j}}{\sum_{i<j} \tilde{\gamma}_{t_i,t_j}} \end{aligned} \tag{18}$$

where γ_{t_i,t_j} denotes the corresponding weight of the IMU–odometer measurements between the ith tag position and the jth tag position. Based on the above weighting function, the IO-MDS cost function can be computed as follows:

$$f(\mathbf{X}_0, \mathbf{X}_1) = \underset{\mathbf{X}_0,\mathbf{X}_1}{\arg\min} \left[\sum_i \rho_{t_i,a_j} \left(d_{t_i,a_j} - r_{t_i,a_j}\right)^2 + \sum_i \gamma_{t_i,t_j} \left(\delta_{t_i,t_j} - l_{t_i,t_j}\right)^2\right] \tag{19}$$

where $l_{t_i,t_j} = \|\mathbf{x}_{t_i} - \mathbf{x}_{t_j}\|_2$ represents the distance between the ith and the jth tag position. The computed coordinates from the IO-MDS algorithm are then aligned with the reference coordinates of the robot trajectory, assuming $\mathbf{x}_{t_1}$ to be the origin.

3) Refined SCAL Based on Factor Graph: We propose a novel algorithm that integrates data from multiple sensors to simultaneously calibrate UWB anchors and localize robots. By leveraging the complementary capabilities of diverse sensors, our algorithm effectively mitigates inherent limitations, resulting in enhanced localization robustness. The proposed approach adopts a factor graph framework to seamlessly fuse information from various sources. A factor graph is a graphical structure used to represent a probabilistic model [30]. In this representation, nodes correspond to random variables, which may encompass observations, concealed states, or other

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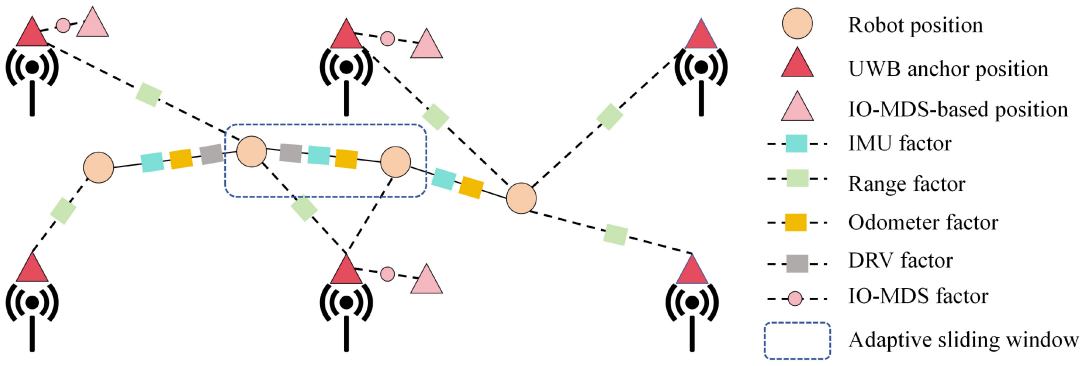


Fig. 2. FGO framework consists of five kinds of factors. The factors of identified LOS ranges from different anchors serve as weak constraints in FGO. To enhance the robustness of position estimation in challenging conditions, we propose the DRV between consecutive ranges and IO-MDS outcomes as strong constraints. Considering that the computation burden of factor graph optimization escalates with the increasing number of factor nodes, we introduce an optimization strategy that confines the optimization variables using an adaptive sliding trajectory window.

unknown parameters. A factor, within this context, is a mathematical function that encapsulates the interdependencies among a set of variables. A factor graph visually illustrates the relationships and constraints among variables by linking nodes to factors through edges. This graph serves as a potent tool for data fusion, enabling the amalgamation of information from diverse sources. Through the introduction of relevant nodes and factors, data from disparate sources are seamlessly integrated into a unified probabilistic model. As depicted in Fig. 2, our framework incorporates IO-MDS, IMU, odometer, range, and DRV factors, which will be elaborated upon in the following sections.

UWB Range Factor: After NLOS identification, we utilize the identified LOS range to establish constraints between the robot position and the UWB anchor position

$$\mathbf{e}_{t_i,a_j}^{\text{Range}} = r_{t_i,a_j} - \left\|\mathbf{x}_{t_i} - \mathbf{x}_{a_j}\right\|_2. \tag{20}$$

IMU Factor: In the UWB–IMU–odometer system, the IMU measurements exhibit high-frequency characteristics, resulting in multiple IMU measurements between two UWB range measurements. Treating each IMU measurement as a separate variable in the factor graph can introduce a significant computational burden during the optimization processes. To avoid this issue, we use preintegration to integrate the IMU measurement between consecutive UWB ranging measurements

$$\mathbf{e}_{t_i}^{\text{IMU}} = \left\|\mathbf{x}_{t_i} - \xi\left(\mathbf{x}_{t_{i-1}}, \eta^b, b^a, b^g\right)\right\|_2 \tag{21}$$

where $\xi(\cdot)$ is the prediction function that represents the ith predicted robot position. Details of IMU preintegration noise propagation and biases update can be found in [31].

Odometer Factor: The odometer effectively mitigates drift and accumulated errors within an IMU-based positioning system through dead reckoning. However, in the factor graph positioning algorithm, the propagation of the position from the odometer measurements fails to meet real-time adjustment requirements for state variables due to asynchronous frequencies of the IMU and odometer measurements. To solve this problem, the odometer factor is incorporated into the factor graph alongside the IMU factor using the preintegration theory [29]. The odometer factor can be defined as

$$\mathbf{e}_{t_i}^{\text{odom}} = \left\|\mathbf{x}_{t_i} - \tau\left(\mathbf{x}_{t_{i-1}}, \eta^o\right)\right\|_2 \tag{22}$$

where $\tau(\cdot)$ denotes the preintegration of odometer measurements.

IO-MDS Factor: This factor is generated from IO-MDS-based position, and it is given as

$$\mathbf{e}_{t_i}^{\text{IO-MDS}} = \left\|\mathbf{x}_{t_i} - \mathbf{x}_{t_i}^{\text{IO-MDS}}\right\|_2 + \left\|\mathbf{x}_{a_j} - \mathbf{x}_{a_j}^{\text{IO-MDS}}\right\|_2 \tag{23}$$

where $\mathbf{x}_{t_i}^{\text{IO-MDS}}$ and $\mathbf{x}_{a_j}^{\text{IO-MDS}}$ are IO-MDS results which are obtained by IO-MDS-based method presented in Section III-B2.

DRV Factor: When the robot moves around the UWB anchor, its range with the anchor changes accordingly. By analyzing this range information, the velocity of a robot along a straight line in 2-D space can be derived using a geometric approach [32]. This velocity information serves as an additional constraint to reduce the impact of IMU/odometer measurement errors on velocity estimation. Inspired by this approach, we introduce a more general radial velocity estimation method which utilizes three instances of range measurements with the same UWB anchor to calculate velocity

$$\begin{bmatrix} r_{t_i,a_j}^2 \\ r_{t_{i+1},a_j}^2 \\ r_{t_{i+2},a_j}^2 \end{bmatrix} = \begin{bmatrix} a^2 + b^2 \\ a^2 + (b + v^r \Delta t_1)^2 \\ a^2 + (b + v^r \Delta t_1 + v^r \Delta t_2)^2 \end{bmatrix} \tag{24}$$

where

$$\Delta t_1 = t_{i+1} - t_i;\ \Delta t_2 = t_{i+2} - t_{i+1}. \tag{25}$$

As shown in Fig 3, the velocity estimate can be obtained by solving the equations

$$v^r = \sqrt{\frac{r_{t_{i+2},a_j}^2 - r_{t_{i+1},a_j}^2 - \frac{\Delta t_2}{\Delta t_1}\left(r_{t_{i+1},a_j}^2 - r_{t_i,a_j}^2\right)}{\Delta t_1 \Delta t_2 + \Delta t_1^2}}. \tag{26}$$

We establish constraints between the velocity calculated from the change in range and the real-time velocity of the robot as follows:

$$e_{t_i}^{\text{V}} = \left\|\mathbf{x}_{t_{i+1}} - \mathbf{x}_{t_i}\right\|_2 - v^r \Delta T \tag{27}$$

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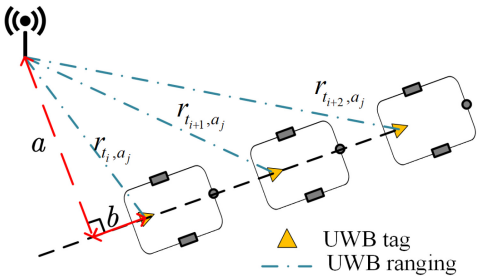


Fig. 3. DRV factor from the consecutive ranging changes. It establishes constraints between adjacent robot translations and enhances the position estimation performance by mitigating the impact of undesired errors on edges within the FGO.

where ΔT is the time interval between the two robot states $x_{t_{i+1}}$ and x_{t_i}.

Optimization Strategy: Considering the escalating computational burden associated with an increasing number of factor nodes in FGO, we propose an adaptive sliding trajectory window to maintain a certain amount of factor nodes from the previous sensor information, thereby ensuring computational tractability for real-time performance. The window length is adaptively updated as follows:

$$L = \begin{cases} L_{\min}, & \alpha \geq \lambda_1 \\ \text{int}\left(\beta \frac{(L_{\max}-L_{\min})(\lambda_1-\alpha)}{\lambda_1-\lambda_0}\right), & \lambda_0 < \alpha < \lambda_1 \\ L_{\max}, & \alpha \leq \lambda_0 \end{cases} \tag{28}$$

where α is the scalar factor which is related to the RSS variance in the length of window $L_{\max}$

$$\alpha = \frac{1}{L_{\max}} \sum_{k=i-L_{\max}+1}^{i} \left(P_k - \frac{\sum_{k=i-L_{\max}+1}^{i} P_k}{L_{\max}} \right)^2. \tag{29}$$

λ_0 and λ_1 denote the upper and lower thresholds of α, $L_{\min}$ and $L_{\max}$ denote the minimum and maximum length values of the sliding window, respectively. int$(\cdot)$ represents the integer function, β is a constant related to the measurement noise frequency.

In summary, the improved factor graph model is constructed by five kinds of factors, including UWB range factor, IMU factor, odometer factor, IO-MDS factor, and DRV factor. Therefore, the optimal state $\widehat{\mathbf{X}} = \{\mathbf{X}_0, \mathbf{X}_1\}$ can be solved as follows:

$$\widehat{\mathbf{X}} = \arg\min \sum_{k=i-L+1}^{i} \left(\left\| \mathbf{e}_{t_k,a_j}^{\text{Range}} \right\|^2_{\Sigma_k^{\text{Range}}} + \left\| \mathbf{e}_{t_i}^{\text{IMU}} \right\|^2_{\Sigma_k^{\text{IMU}}} + \left\| \mathbf{e}_{t_i}^{\text{odom}} \right\|^2_{\Sigma_k^{\text{odom}}} + \left\| \mathbf{e}_{t_i,a_j}^{\text{IO-MDS}} \right\|^2_{\Sigma_k^{\text{IO-MDS}}} + \left\| \mathbf{e}_{t_i}^{\text{V}} \right\|^2_{\Sigma_k^{\text{V}}} \right) \tag{30}$$

where Σ_k^{Range}, Σ_k^{IMU}, Σ_k^{odom}, $\Sigma_k^{\text{IO-MDS}}$, and Σ_k^{V} are the covariance matrix associated with UWB range, IMU, odometer measurements, and IO-MDS and DRV results, respectively.

For the real-time performance of the system, a termination criterion is introduced to finish the UWB anchor position optimization process as the following condition:

$$\max\{\sigma_{a_j}\} < \sigma_a, j \in [1, N] \tag{31}$$

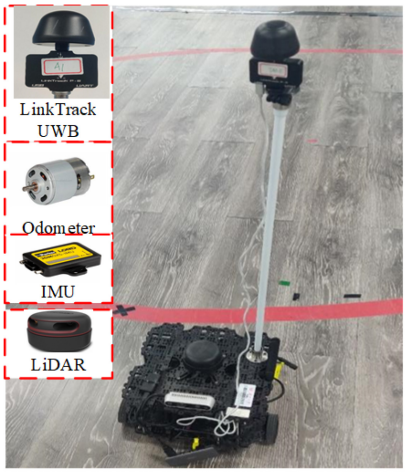


Fig. 4. Experimental platform.

where σ_{a_j} is the sample variance of the jth anchor's positions σ_a is the preset threshold and N is the number of anchors. Once the criterion is met, the anchor position estimate is considered fixed.

IV. Experiments

A. Experimental Setup

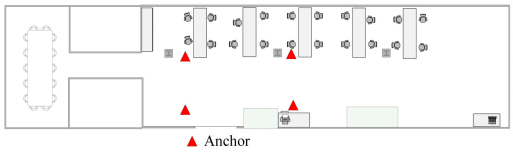
We evaluate the performance of the proposed LOS/NLOS identification and simultaneous calibration and localization (SCAL) algorithms on the Turtlebot3 platform, which is equipped with a UWB tag, namely, Nooploop LinkTrack, IMU, and odometer, as shown in Fig. 4. The UWB system has been configured to transmit ranging and RSS data between UWB anchors and a tag, with a maximum effective communication distance of 100 m. However, the data transmission interface between anchors is not accessible to users. Only the tag and anchors support two-way data transmission to facilitate communication. The measurement rate for each UWB anchor is set to 32 Hz, while odometer and IMU measurements are recorded at 28 Hz and 127 Hz, respectively. Both the sensor data acquisition software and our algorithms operate on the CPU of ARMv8 (NVIDIA Jetson TX2), with a built-in Melodic robot operating system (ROS).

Experimental campaigns are conducted in various test scenarios, as shown in Figs. 5 and 6. Fig. 5 represents the laboratory of 9.8 m×37 m, including a whiteboard, glass wall, and 40-cm-thick concrete columns. Fig. 6 is the underground parking lot of 63.2 m×35.2 m, including cars and 70-cm-thick concrete columns.

B. LOS/NLOS Identification Performance

1) Model Training: We collected UWB measurements, including the raw ranging distances and RSS, to train the LS-SVM model for both LOS and NLOS cases. The horizontal distance between the tag and anchor ranged from 1 m to 20 m, with an increment of 1 m for LOS cases. During the NLOS data collection phase, the tag moved incrementally by 0.5 m away from the fixed anchor, as illustrated in Table III. Both LOS and NLOS data were labeled and recorded in CSV

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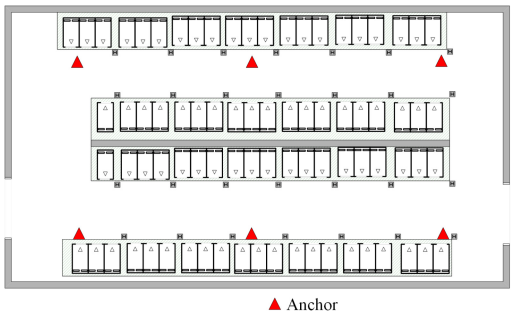


(a)

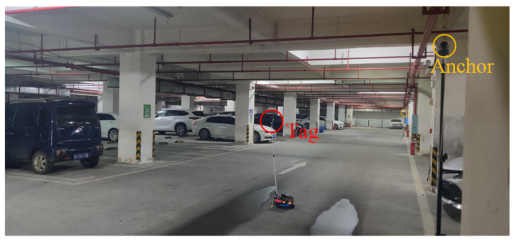


(b)

Fig. 5. Laboratory experimental environment. (a) Deployment of anchors in the lab. (b) Setup of the UWB anchors and the tag.



(a)



(b)

Fig. 6. Parking lot experimental environment. (a) Deployment of anchors in the parking lot. (b) Setup of the UWB anchors and the tag.

files, totaling 7500 records. The laser rangefinder provided the actual distance at each sample point.

2) Identification Accuracy: We use the following three common metrics (recall, precision, and accuracy) to evaluate the identification accuracy:

$$\text{Recall} = \frac{\text{TP}}{\text{TP}+\text{FN}} \tag{32}$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP}+\text{FP}}$$

$$\text{Accuracy} = \frac{\text{TP}+\text{TN}}{\text{TP}+\text{TN}+\text{FP}+\text{FN}}$$

TABLE III
DISTANCE BETWEEN THE ANCHOR AND THE TAG IN TWO SCENARIOS

Scene	LOS/NLOS	Obstacle	Distance
Laboratory	LOS	-	[1, 20 m]
	NLOS	Whiteboard	[0.5, 5.5 m]
	NLOS	Glass wall	[1, 10 m]
	NLOS	Concrete column	[0.5, 4 m]
Parking lot	LOS	-	[1, 20 m]
	NLOS	Car	[0.5, 5.5 m]
	NLOS	Concrete column	[0.5, 4 m]

TABLE IV
LOS/NLOS IDENTIFICATION ACCURACY IN THE LAB DATASET

Method	Recall	Precision	Accuracy
LS-SVM [8]	93.0%	94.3%	93.4%
RF [21]	89.2%	89.1%	89.6%
SVM [23]	90.0%	89.2%	89.0%
ILS-SVM (RSS only)	93.7%	93.2%	93.4%
ILS-SVM (ranging only)	94.0%	93.8%	93.5%
ILS-SVM	**94.5%**	**94.6%**	**94.5%**

TABLE V
LOS/NLOS IDENTIFICATION ACCURACY IN THE PARKING LOT DATASET

Method	Recall	Precision	Accuracy
LS-SVM [8]	94.4%	95.2%	94.0%
RF [21]	91.2%	89.5%	90.4%
SVM [23]	92.6%	91.2%	92.9%
ILS-SVM (RSS only)	94.6%	95.5%	94.3%
ILS-SVM (ranging only)	94.2%	95.2%	94.2%
ILS-SVM	**95.0%**	**95.8%**	**95.4%**

where true positive (TP)/true negative (TN) is the number of data correctly identified as LOS/NLOS, false positive (FP)/false negative (FN) indicates falsely identified as LOS/NLOS, respectively. We randomly split 30% data from the collected data for testing the proposed ILS-SVM, LS-SVM [8], RF classifier [21], and SVM [23]. The experimental results are illustrated in Tables IV and V.

Our proposed algorithm demonstrates the best mean identification recall (94.75%), precision (95.20%), and accuracy (94.95%), in both laboratory and parking lot scenarios, surpassing the LS-SVM [8], RF classifier [21], and SVM [23]. As shown in Tables IV and V, the identification accuracy in the parking lot exceeds that in the laboratory. This can be attributed to the increased complexity of the laboratory environment, with more obstacles near the anchors causing multipath signal effects and thereby reducing identification accuracy. With the help of the multidimensional characteristics of UWB signals combined with the adaptive weight parameter, our proposed ILS-SVM method exhibits high robustness in identification. Besides, the combination of multikernel functions enhances the robustness of classical LS-SVM [8], ensuring minimal impact on identification results even in the presence of UWB signal noise. We also use RSS and ranging samples separately as input features for the proposed ILS-SVM, respectively. Tables IV and V show that, by combining

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TABLE VI
ERRORS (IN METERS) OF THE ROBOT LOCALIZATION AND ANCHOR SELF-CALIBRATION IN THE LAB DATASET. *The Bold* DENOTES THE BEST LOCALIZATION PERFORMANCE

Method	Robot Localization (ATE)						Anchor Self-calibration (APE)					
	RMSE	Mean	Median	Std	Min	Max	RMSE	Mean	Median	Std	Min	Max
ESKF [15]	0.153	0.138	0.136	0.067	0.009	0.327	0.436	0.408	0.415	0.155	0.184	0.617
PF [16]	0.183	0.168	0.164	0.073	0.024	0.320	0.512	0.483	0.373	0.170	0.262	0.725
MGBP [17]	0.103	0.091	0.075	0.048	0.014	0.283	0.357	0.331	0.373	0.133	0.111	0.467
IPFHO [18]	0.152	0.133	0.114	0.073	**0.006**	0.301	0.417	0.367	0.365	0.199	**0.101**	0.637
VB-CSRUKF-RTS [19]	0.146	0.131	0.113	0.065	0.014	0.317	0.396	0.341	0.300	0.201	0.148	0.617
Ours	**0.059**	**0.052**	**0.049**	**0.028**	0.011	**0.282**	**0.203**	**0.196**	**0.202**	**0.054**	0.114	**0.263**

these two types of measurements, our method reduces the uncertainty associated with each measurement type, leading to more accurate classification of LOS and NLOS conditions compared to ILS-SVM methods that use only one type of signal (either RSS or ranging).

C. SCAL Evaluation

With the same experimental setup, we further performed self-calibration and robot localization experiments across different sizes of deployment areas, comparing our proposed method with the five reference methods ESKF [15], PF [16], MGBP [17], IPFHO [18] and VB-CSRUKF-RTS [19] that are described in Section II. UWB anchors with known height are placed at unknown locations, the mobile robot moves a random trajectory in the environment to simultaneously locate the anchors and track its own position. The ground truth of the robot path is tracked by the SLAM solution named LaMa2[3] based on the LIDAR and odometer installed on the turtlebot3. Anchor locations are measured using a laser range finder to ensure high-accuracy ground truth values.

The basic performance metrics for our method are absolute pose error (APE) and absolute trajectory error (ATE).

APE: This metric is used to calculate the difference of position changes between the estimated and ground truth anchor position. The root mean square position error (RMSE) of APE is defined as

$$\mathrm{APE}_{\mathrm{RMSE}} = \sqrt{\frac{1}{N}\sum_{n=1}^{N}\|\widehat{\mathbf{a}}_n - \mathbf{a}_n\|_2^2} \tag{33}$$

where $\widehat{\mathbf{a}}_n$ is the nth estimated anchor position and a_n is its ground truth position. N is the total number of anchors.

ATE: This metric is used to calculate a deviation after the overall alignment between the estimated robot trajectory and the ground truth trajectory. The RMSE of ATE is calculated as

$$ATE_{\mathrm{RMSE}} = \sqrt{\frac{1}{M}\sum_{i=1}^{M}\left\|\mathrm{trans}\left(F_i^{-1}S\hat{F}_i\right)\right\|_2^2} \tag{34}$$

where S is the yaw-only transformation corresponding to the least-squares solution that aligns the estimated trajectory $\hat{F}_{1:M}$ with the ground truth trajectory $F_{1:M}$. $trans(\cdot)$ denotes the translational components of the transformation matrix, M is the total number of poses in the trajectory sequence.

[3]https://github.com/iris-ua/iris_lama

We also select Mean, Std, Median, Minimum, and Maximum of APE and ATE as the performance metrics to separately evaluate anchor calibration and robot localization.

1) Laboratory: In the laboratory, we adopt a layout design consisting of four anchors and one tag to evaluate SCAL methods. Two anchors are obstructed by obstacles to create LOS/NLOS mixed conditions. Fig. 7 shows the anchor calibration results and the estimated trajectory obtained by our proposed method alongside five reference methods. The red circles denote the ground truth positions of the anchors, while the red line represents the ground truth trajectory. The green line depicts the estimated trajectory. It is evident from the results that the localization trajectory of the mobile robot and anchor locations solved by our proposed method closely align with the ground truth.

Our method provides a good initial guess of the anchor position during the initialization step, facilitating the convergence of localization results. In the second calibration stage, IO-MDS and DRV factors provide robust constraints within the solution space for SCAL estimation. Thus, our method yields accurate estimations even in NLOS conditions.

The errors in the robot localization and anchor self-calibration produced by the six methods are detailed in Table VI. Regarding the RMSE, we observe that our proposed method yields trajectory deviations of 0.094 m, 0.124 m, 0.044 m 0.093 m and 0.087 m lower than those of ESKF, PF, MGBP, IPFHO and VB-CSRUKF-RTS, respectively, in the laboratory scenario. Moreover, the RMSE of estimated anchor locations is only 0.203 m, much lower than ESKF (0.436 m), PF (0.512 m), MGBP (0.357 m), IPFHO (0.417 m) and VB-CSRUKF-RTS (0.396 m). These metrics verify that our proposed method, based on the factor graph, refines both anchor and robot positioning, enhancing the accuracy and robustness of trajectory and calibration results.

2) Parking Lot: In the parking lot, we adopt the layout design of six anchors and one tag to evaluate the accuracy of six methods. Fig. 8 shows the robot and anchor self-calibration results obtained by ESKF, PF, MGBP, IPFHO, VB-CSRUKF-RTS, and our proposed method over the entire trajectory. Similar to the laboratory scenario, our proposed method achieves more consistency with the ground truth in both robot and anchor position estimations.

Table VII compares the detailed errors in robot localization and anchor self-calibration generated by the six methods. Our proposed method outperforms all other methods in both robot and anchor location estimation. For the accuracy of

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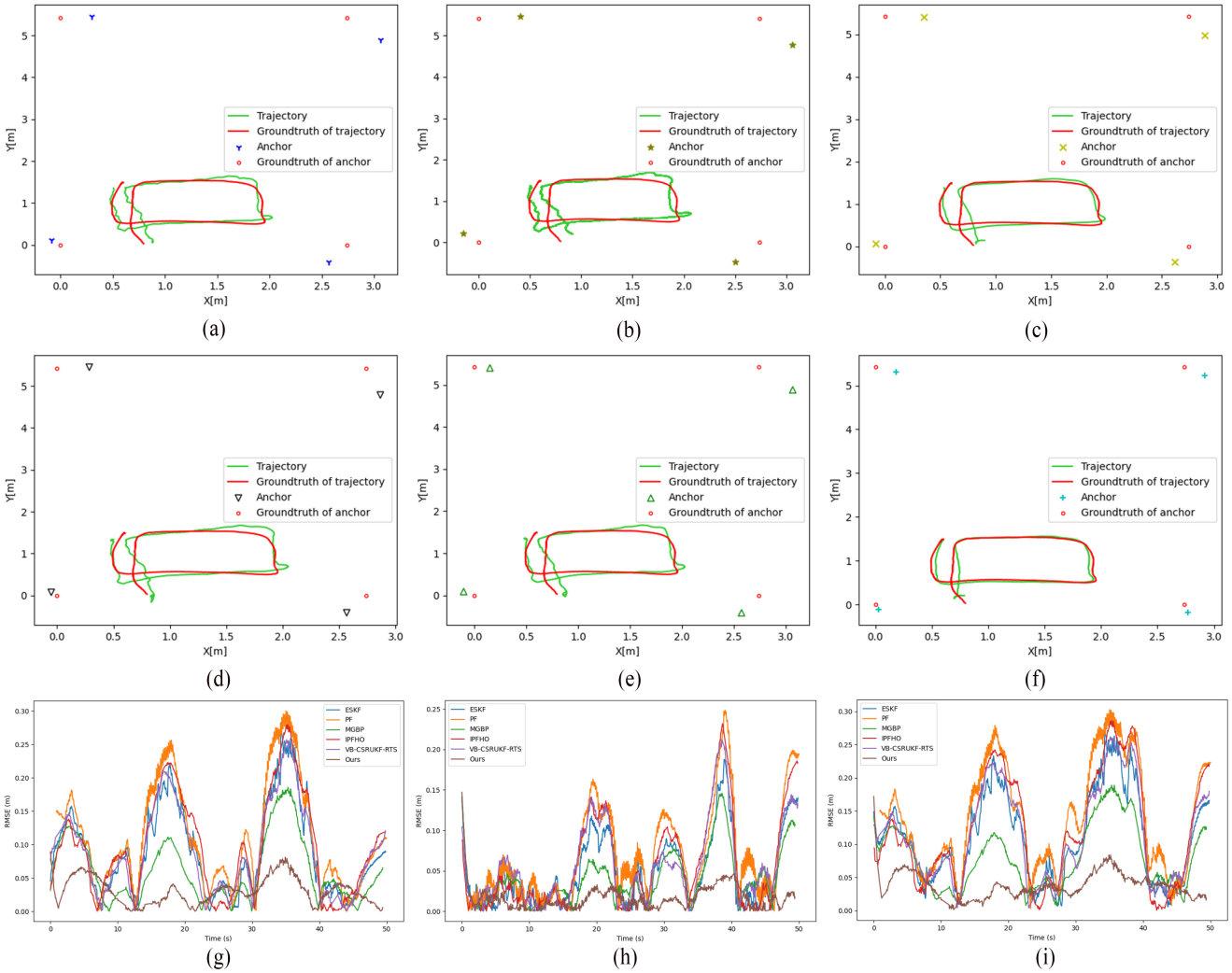


Fig. 7. Trajectory and calibration results comparison for ESKF, PF, MGBP, IPFHO, VB-CSRUKF-RTS, and the proposed method in the lab environment. The proposed method allows the joint optimization of robot and anchor position estimations to converge to more accurate results than all other methods. (a) ESKF. (b) PF. (c) MGBP. (d) IPFHO. (e) VB-CSRUKF-RTS. (f) Ours. (g) RMSE in X-axis. (h) RMSE in Y-axis. (i) RMSE.

TABLE VII
ERRORS (IN METERS) OF THE ROBOT LOCALIZATION AND ANCHOR SELF-CALIBRATION IN THE PARKING LOT DATASET. *The Bold* DENOTES THE BEST LOCALIZATION PERFORMANCE

Method	Robot Localization (ATE)						Anchor Self-calibration (APE)					
	RMSE	Mean	Median	Std	Min	Max	RMSE	Mean	Median	Std	Min	Max
ESKF [15]	0.718	0.670	0.669	0.257	0.030	2.094	1.080	1.014	1.099	0.372	0.516	1.428
PF [16]	0.730	0.692	0.701	0.231	0.069	1.589	0.947	0.889	1.087	0.328	0.342	1.156
MGBP [17]	0.602	0.565	0.584	0.208	**0.008**	1.468	0.851	0.787	0.792	0.325	0.245	1.247
IPFHO [18]	0.681	0.639	0.654	0.235	0.022	1.534	0.891	0.821	**0.696**	0.346	0.516	1.511
VB-CSRUKF-RTS [19]	0.671	0.630	0.643	0.231	0.021	1.500	0.792	0.739	0.730	**0.285**	0.245	1.086
Ours	**0.507**	**0.466**	**0.436**	**0.201**	0.018	**1.291**	**0.763**	**0.701**	0.748	0.302	**0.101**	**1.047**

robot positioning, our method achieves an RMSE of 0.507 m, better than ESKF (0.718 m), PF (0.730 m), MGBP (0.602 m), IPFHO (0.681 m) and VB-CSRUKF-RTS (0.671 m). Due to the longer trajectory length in the parking lot compared to the laboratory setup, the accumulated errors of the robot trajectory are more pronounced. Regarding anchor self-calibration accuracy, our method achieves an RMSE of 0.763 m, better than ESKF (1.080 m), PF (0.947 m), MGBP (0.851 m), IPFHO (0.891 m), and VB-CSRUKF-RTS (0.792 m). In conclusion, our method demonstrates satisfactory performance even in the large-scale deployment scenario, which holds significant importance for practical applications.

Good initial guesses play a crucial role in achieving fast convergence and robustness in self-calibration processes. ESKF adopts a loosely coupled fusion method that gathers UWB range measurements and IMU propagated positions to calculate the initial anchor positions via the multilateration equations. However, this method struggles to handle NLOS range errors and IMU measurement noises, often resulting in biased estimates. PF and IPFHO introduce a region-based

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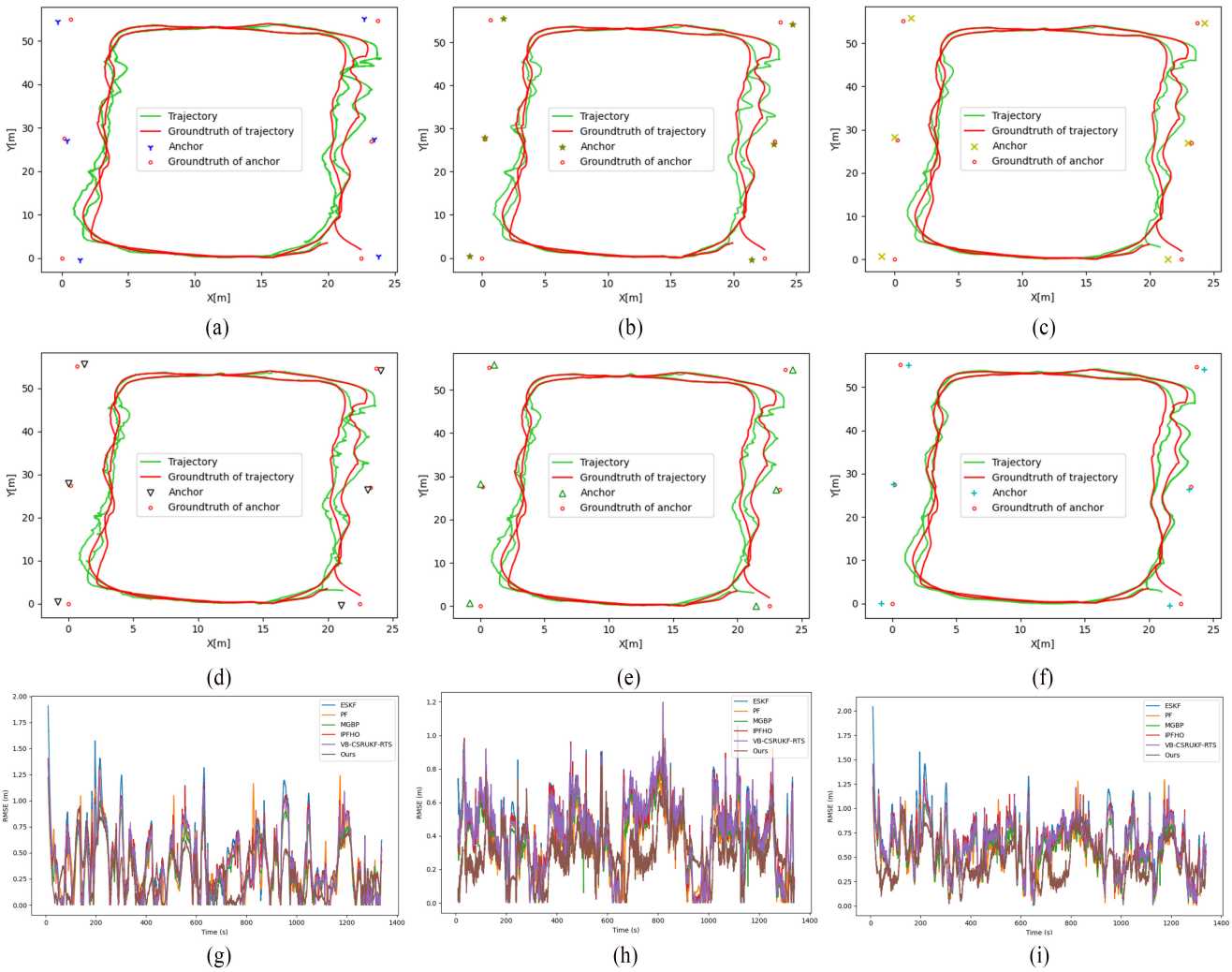


Fig. 8. Trajectory and calibration results for ESKF, PF, MGBP, IPFHO, VB-CSRUKF-RTS, and the proposed method in the parking lot environment. Although the robot trajectory of the proposed method occasionally has several slightly obvious trajectory deviations, the overall accuracy is still better than other comparison methods. (a) ESKF. (b) PF. (c) MGBP. (d) IPFHO. (e) VB-CSRUKF-RTS. (f) Ours (g) RMSE in X-axis. (h) RMSE in Y-axis. (i) RMSE.

particle filter and adaptive particle filter, respectively, for anchor location initialization with low-computational cost. However, these methods' effectiveness is hindered by the convergence rate of particles. Due to the relatively low precision of particle initialization, especially when dealing with the extensive scales of state space during anchor initialization, PF and IPFHO exhibit significant location deviations, particularly noticeable in parking lot environments. MGBP does not require an additional estimator during the initialization process, which shows a lower calculated amount when compared with the existing graph optimization methods. However, MGBP relies solely on ranging data for NLOS detection through a hypothesis test, which may compromise its accuracy in NLOS scenarios. VB-CSRUKF-RTS integrates multiple optimization algorithms to improve the UWB positioning, however, it relies on RNs under LOS condition.

D. Ablation Study

In the first coarse initialization stage, we introduced a dynamic weight to adjust the contributions of UWB ranging and IMU–odometer measurements in the cost function based on observation noises. To evaluate its effectiveness, we conducted ablation experiments on both lab and parking lot datasets, comparing the performance of the system with the dynamic weight to that of a baseline model without this feature.

Table VIII indicates that the introduction of the dynamic weight reduced anchor position drift, thus ensuring geometric accuracy in anchor calibration and ultimately improving overall robot localization performance.

Additionally, to verify the effects of the proposed IO-MDS factor and DRV factor, we performed further ablation experiments on the same datasets, comparing our method without the DRV factor (Ours w/o DRV factor) and without IO-MDS factor (Ours w/o IO-MDS factor). Table IX presents the ATE errors for these three methods. Both the DRV and IO-MDS factors reduce the pose estimation error and improve our robustness by narrowing the error distribution.

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TABLE VIII
ABLATION STUDY RESULTS (APE IN METER) OF ANCHOR SELF-CALIBRATION IN BOTH LAB AND PARKING LOT DATASET. *The Bold* DENOTES THE BEST LOCALIZATION PERFORMANCE

Method	Lab dataset						Parking lot dataset					
	RMSE	Mean	Median	Std	Min	Max	RMSE	Mean	Median	Std	Min	Max
Ours w/o Adaptive weight	0.411	0.366	0.365	0.186	0.117	0.617	0.894	0.854	0.915	0.263	0.382	1.191
Ours	**0.203**	**0.196**	**0.202**	**0.054**	**0.114**	**0.263**	**0.763**	**0.701**	**0.748**	**0.302**	**0.101**	**1.047**

TABLE IX
ABLATION STUDY RESULTS (ATE IN METER) OF THE ROBOT LOCALIZATION IN BOTH LAB AND PARKING LOT DATASET. *The Bold* DENOTES THE BEST LOCALIZATION PERFORMANCE

Method	Lab dataset						Parking lot dataset					
	RMSE	Mean	Median	Std	Min	Max	RMSE	Mean	Median	Std	Min	Max
Ours w/o DRV factor	0.149	0.126	0.104	0.078	0.027	0.360	0.687	0.631	0.625	0.270	**0.011**	1.658
Ours w/o IO-MDS factor	0.147	0.125	0.104	0.076	0.024	0.363	0.663	0.616	0.623	0.245	0.019	1.725
Ours	**0.059**	**0.052**	**0.049**	**0.028**	**0.011**	**0.282**	**0.507**	**0.466**	**0.436**	**0.201**	0.018	**1.291**

V. CONCLUSION

This article presents a novel SCAL algorithm framework that integrates UWB, IMU, and odometer data to perform global optimization on the positioning of robot trajectories and anchors in LOS/NLOS mixed environments. We propose an ILS-SVM for NLOS identification and a two-stage calibration scheme for both anchor self-calibration and robot localization, without the need for anchor-to-anchor communication or known initial values for anchor locations. Experimental results show that our method outperforms five reference methods in both laboratory and parking lot datasets. In the future, we plan to incorporate both LOS and NLOS measurements in a hybrid approach and extend our work to include multiple moving robots with automatic configuration of UWB nodes.

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